



Decentralizing Agriculture

A shift in the industrial paradigm of food production

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Contents

1	Introduction	4
1.1	Motivation	4
1.2	Project vision	4
2	The problems of industrial agriculture	5
2.1	Land Use	5
2.2	Biodiversity loss	6
2.3	Soil degradation and erosion	6
2.4	Fertilizer use	7
2.5	Pesticide use	8
2.6	Monoculture	8
2.7	Freshwater depletion	9
2.8	Greenhouse gas emissions	9
2.9	Produce transportation, storage and disposal	9
2.10	The existential dependency on a few producers	9
2.11	Suppression of dietary diversity and nutritional quality	10
3	Background: The Netherlands as a Proof of Concept	11
3.1	The Dutch agricultural model	11
3.2	The Triple Helix institutional model	11
3.3	Limitations and honest framing	12
3.4	Relevance to the proposed system	12
4	The proposed solution	13
4.1	Concept overview	13
4.2	Alignment with identified problems	13
4.3	Scope and niche	14
5	Nutritional self-sufficiency analysis	16
5.1	The calorie problem	16
5.2	The correct framing: nutritional self-sufficiency	16
5.3	Vertical stacking multiplier	16
6	Limitations and mitigations	17
6.1	Energy intensity	17
6.2	Cold-start problem for the ML model	17
6.3	Open-source hardware complexity	17
6.4	Data governance and quality	18

7	System architecture	19
7.1	Overview	19
7.2	The greenhouse node: MCU/SBC split	19
7.3	The data flywheel	20
7.4	Offline-first principle	20
8	Technology stack	21
8.1	Embedded firmware (MCU)	21
8.2	Edge gateway (SBC, Linux)	22
8.3	Hardware support tiers	22
8.4	Cloud / data commons	23
8.5	Licences	23
9	The data schema	24
9.1	Sensor telemetry record	24
9.2	Image record	25
10	The climate-aware species recommender	25
10.1	Concept	25
10.2	Input variables	25
10.3	Output	25
10.4	Why this is not an ML model	26
10.5	Climate adaptation over time	26
11	Open-source strategy	27
11.1	The data commons as the core asset	27
11.2	Community governance	27
11.3	Self-hostability	27
12	Impact assessment	27
13	Future work	28
	References	29

Abstract

Industrial agriculture is responsible for the majority of global biodiversity loss, roughly one third of all greenhouse gas emissions, and the creation of fragile, highly concentrated food supply chains on which billions of people existentially depend. This paper proposes an open-source controlled environment agriculture (CEA) platform designed to address these problems through decentralisation, automation, and a shared data commons.

The system comprises a sensor array and camera mounted in a small-scale greenhouse, a microcontroller running a PID control loop over environmental actuators, a single-board computer performing edge ML inference, and an optional connection to a cloud-based data flywheel where growth models are continuously retrained on anonymised telemetry from the global network of deployed nodes. A computer vision pipeline provides growth metrics, disease diagnosis, and species recognition. A climate-aware species recommendation engine minimises energy consumption by matching recommended crops to the local ambient environment. All hardware designs, firmware, software, trained models, and collected data are released under open licences, enabling a virtuous cycle: the more nodes deployed, the richer the shared dataset, and the more accurate the growth optimisation.

The project does not aim to replace field agriculture wholesale. Its niche is high-value, perishable, water-intensive crops grown in proximity to consumers — the segment where the environmental and supply-chain benefits of CEA are largest and most defensible. Within that niche, we argue the platform can meaningfully reduce land use pressure, water consumption, fertiliser runoff, pesticide pollution, transport emissions, food waste, and supply-chain concentration, while enabling a diversity of crops that the industrial monoculture model structurally cannot support.

1 Introduction

1.1 Motivation

The global food system is in a state of structural crisis. Industrial agricultural production — the dominant paradigm for the past century — has delivered extraordinary increases in caloric output, but at costs that are becoming impossible to ignore: ecosystem collapse, soil exhaustion, water depletion, climate destabilisation, and the creation of food supply chains so concentrated that a single geopolitical shock can threaten the nutrition of hundreds of millions of people.

At the same time, the tools to build something better have never been more accessible. Capable micro-controllers cost a few dollars. Machine learning frameworks are open source. Camera modules capable of running computer vision inference are commodity hardware. The question is no longer whether small-scale, intelligent, automated food production is technically feasible — it is — but whether it can be made cheap enough, robust enough, and open enough to displace the incumbent system at meaningful scale.

This paper describes a platform designed to make that possible.

1.2 Project vision

The project’s core proposition is that a network of many small, locally operated, data-connected greenhouses can collectively outperform a system of few large, centralised farms — on environmental metrics, on resilience, and eventually on cost — provided that the intelligence required to operate them efficiently is shared as a common good rather than locked up in proprietary systems.

The platform is designed around four interlocking principles:

1. **Edge-first control.** All critical control logic runs locally and offline. The system operates without internet connectivity. The cloud connection is optional and asynchronous, used only for learning and coordination, never for real-time operation.
2. **Open everything.** Hardware reference designs, firmware, software, trained models, and the collected dataset are all released under maximally permissive open licences. Anyone can deploy, modify, and contribute back.
3. **Data flywheel.** Every deployed node, with the user’s consent, contributes anonymised telemetry to a shared dataset. Models trained on this dataset are continuously improved and pushed back to all nodes via over-the-air updates. The network becomes smarter as it grows.
4. **Climate-adapted intelligence.** The system does not prescribe a fixed set of crops. A climate-aware recommendation engine suggests species whose optimal growing conditions most closely match the local ambient environment, minimising the energy required to close the gap between what the environment provides and what each crop needs.

2 The problems of industrial agriculture

Although the rise of the modern industrial structure of agricultural production has given many societies the ability to forestall famine and thrive in ways long thought impossible, many of its social and ecological repercussions have revealed themselves to be potentially dangerous in the best scenario and significant contributors to an existential crisis in the worst. Every step throughout the production process results in ecological problems of varying severity, from the clearing of the land to the delivery of the final product, each impacting the surrounding ecosystems and underlying soils in different ways. It is a production model that forces the whole population into the dependence of a few producers, who have increasingly incorporated more and more monoculture practices, thus making the food supply more vulnerable to collapse in the case of an uncontrolled pest. The following sections document each of these problems in turn, drawing on recent peer-reviewed literature and institutional reports.

2.1 Land Use

[SOURCES CHECKED]

The prevailing paradigm of industrial agricultural production requires the use of vast swathes of land, along with a considerable amount of water, energy and other resources, which inevitably ends up homogenizing the local flora and destroying the habitat of the local fauna, leading to an overall loss of biodiversity. The agricultural expansion that the current system entails is indeed one of the main drivers of global biodiversity loss [1, p. 412], [2, p. 199], and is the most widespread form of land-use change, with cropping and animal husbandry covering over one third of the terrestrial land surface [1, p. XVI]. More than 90% of global biodiversity impacts from land-use change are attributable to agriculture, with crop cultivation accounting for 72% and pastures for 21% — dwarfing the combined impact of mining, urban development, and all other industries [3].

This geographical expansion of the agricultural sector has been a constant for decades, having grown by 78 million ha (780,000 km²) between 2001 and 2023, mostly driven by an increase in permanent crops such as cocoa, oil palm or coffee [4]. All of these are used intensively for monoculture, as are feedstock crops which by themselves account for one third of the total crops produced globally [5, p. 22]; this particular form of agriculture, which consists in the the use of a single crop for large productions, is another potent driver of biodiversity and habitat loss which will be explored in a subsequent section.

The extension of anthropogenic drivers of change has led to as little as 13% of the ocean and 23% of the land falling under the classification of ‘wilderness’. These few remaining areas that lie still outside of humanity’s sphere of influence tend to be remote or unproductive, like tundras or oceanic gyres, the most hospitable biomes having been almost or completely modified by humans in most regions [1, p. 206]. With the transformation for human convenience of the landscapes of the world that are the most accommodating to life, displaced wild populations are left without a habitat.

The impact of this intrusion into the wilderness has also far-reaching consequences as well. According to the FAO’s State of Food and Agriculture report [6, p. 4], agriculture currently drives 90 percent of global deforestation, with cropland expansion accounting for almost half of the total deforested area. This leads

to changes in both global and local climates through two main mechanisms: the release of stored carbon produces a robust warming signal worldwide, while the removal of forest cover changes surface albedo, evapotranspiration and canopy roughness, which produces a global warming effect for deforestation in the tropics, and a cooling effect at higher latitudes. At the local scale, forest loss almost always raises surface temperatures, with typical increases between 0.2 °C - 2 °C, depending on latitude [7].

A pan-tropical synthesis of deforestation data found that between 90 and 99% of all tropical deforestation between 2011 and 2015 was directly or indirectly driven by agricultural expansion, with 6.4–8.8 million hectares cleared per year [8]. Crucially, only 45–65% of that deforested land was subsequently converted into productive agricultural use, meaning that a substantial fraction of tropical forest destruction delivers no lasting food-production benefit.

2.2 Biodiversity loss

[SOURCES CHECKED]

Beyond direct habitat destruction, the intensification of agriculture has caused cascading losses across all taxa. The spread and intensification of agriculture over the past half century — through greatly expanded scales, monoculturing, increasing pesticide and fertiliser application, and the elimination of hedgerows and other wildlife habitat — is directly related to major documented declines in insect biomass and diversity [9]. At the global scale, agriculture alone has been identified as a threat to 24,000 of the 28,000 (86%) species currently at risk of extinction [10].

The impact reaches beyond field boundaries. A 27-year study using standardised traps across 63 German nature protection areas found a seasonal decline of 76% and a mid-summer peak decline of 82% in total flying insect biomass, with losses apparent regardless of habitat type and implying landscape-scale drivers rooted in agricultural intensification [11]. This decline is multi-causal and compounding: habitat loss, agricultural conversion, climate change, and pesticide exposure co-occur and interact in intensively farmed landscapes, with annual abundance declines frequently around 1–2% across taxa [12]. Across a systematic review of 73 reports, approximately 41% of insect species are declining and a third are threatened with extinction; the pace of local insect species extinction is eight times higher than that of vertebrates, with total insect biomass estimated to be declining at approximately 2.5% per year globally. [13].

2.3 Soil degradation and erosion

[SOURCES CHECKED]

Soils underpin the production of ecosystem goods and services that sustain both natural habitats and human societies. They host the full spectrum of biodiversity, from bacteria and fungi to plants and macro-fauna; and this biotic community drives the soil functions (nutrient cycling, water regulation, carbon sequestration, etc.) that generate those services. Those functions are controlled primarily by the soil’s chemical composition, biological activity, and physical structure. When any of these attributes deteriorate, the soil’s ability to deliver essential ecosystem services, such as food production, water purification, and climate regulation, is compromised or lost [14, pp. 4–5].

Industrial tillage-based agriculture destroys soil at a rate that outpaces natural soil formation by orders of magnitude. Erosion rates from conventionally ploughed agricultural fields average one to two orders of magnitude greater than rates of soil production, erosion under native vegetation, and long-term geological erosion — making conventional plow-based agriculture fundamentally unsustainable over civilizational timescales [15]. Industrial farming practices such as mono-cropping, use of organic fertilizers, and removal of crop residues are a primary cause of the decline in soil organic matter, which is essential to long-term soil productivity [16].

The IPCC’s Special Report on Climate Change and Land estimates that cropland soils have lost 20–60% of their organic carbon content relative to pre-cultivation states, and that soils under conventional agriculture continue to act as a net source of greenhouse gas emissions [17].

A second-order meta-analysis synthesizing 230 first-order meta-analyses encompassing more than 25,000 primary studies confirms that land conversion for crop production is a consistently net-negative driver for soil organic carbon (SOC), with mean SOC losses of approximately 25% following conversion from forest or wetland, and 16% from grassland, though most studies underestimate actual losses because they measure only over years to decades, whereas full equilibration after land-use change may take around 80 years. This matters because SOC is foundational to soil health, food production, and climate regulation: global SOC stocks to 2 m depth are estimated at approximately 2,400 Gt C — three times the carbon currently held in the atmosphere — meaning even small proportional losses translate into significant atmospheric CO₂ increases. SOC restoration is possible, but it is understood to be generally slower than depletion [18].

2.4 Fertilizer use

Excess nitrogen and phosphorus from agricultural fertilizers leach into waterways, triggering eutrophication — a process in which accelerated algal growth reduces water clarity and, when these blooms die, microbial decomposition severely depletes dissolved oxygen, creating hypoxic or anoxic dead zones that eliminate habitat for aquatic invertebrates, fish, and other organisms, driving localized biodiversity collapse. Cultural eutrophication of this kind has been identified as a leading cause of freshwater and coastal ecosystem impairment worldwide. [19], [20]

Nearly 50% or more of applied nitrogen is lost to the environment through various mechanisms [21], [22]. Eutrophication caused by over-enrichment with phosphorus and nitrogen is a widespread problem in rivers, lakes, estuaries, and coastal waters, leading to toxic algal blooms, oxygen loss, fish kills, loss of seagrass beds, and biodiversity loss [23]. [TODO: this is a bit redundant, rephrase]

[Checked sources until this point]

Excess nitrogen in agricultural soils undergoes microbial conversion to nitrous oxide (N₂O) — a greenhouse gas with a global warming potential 273 times that of CO₂ over 100 years [24] and the leading ozone-depleting substance currently emitted by human activities [25]. Global anthropogenic N₂O emissions increased by 40% between 1980 and 2020, driven primarily by agricultural nitrogen use, and atmospheric concentrations have now exceeded all IPCC-projected scenarios, threatening Paris Agreement targets [26].

Critically, the relationship between nitrogen application and N₂O emissions is not linear but exponential: each additional unit of fertiliser applied above crop demand generates a disproportionately larger emission

penalty, meaning that small reductions in over-application can produce outsized environmental gains [27].

2.5 Pesticide use

Pesticides contaminate soil, water, and air well beyond the target application zone, harming non-target organisms throughout the food web. Continuous and excessive pesticide use leads to the accumulation of toxic chemicals in soil, harming beneficial organisms such as earthworms and soil microorganisms, disrupting soil fertility and overall ecosystem health [28]. 64% of the world’s agricultural land — covering approximately 24.5 million km² — has been affected by pesticide use, and water resources in countries including South Africa, China, India, Australia, and Argentina are heavily contaminated with pesticide residues.

Among the most consequential sub-effects of pesticide use is its documented role in the collapse of pollinator populations. Neonicotinoid insecticides, the most widely used insecticide class globally, cause a range of sublethal effects in bee colonies — including impaired foraging, damaged navigation, suppressed immune function, and reduced reproductive success — that interact synergistically with pathogen loads to produce colony collapse [29]. A 23-year survey across 2.8 million km² identified neonicotinoid compounds as the pesticide class most strongly associated with the precipitous decline of the western bumble bee (*Bombus occidentalis*), with further acceleration projected under most future scenarios [30]. This is a critical feedback for food security: approximately 75% of cultivated food plants depend on insect pollination, meaning that chemical impairment of pollinators ultimately undermines the very productivity that pesticide applications are intended to protect.

2.6 Monoculture

Growing the same crop on the same land year after year degrades soil, strips biodiversity, and makes the entire food system vulnerable to disease and climate shocks. A fundamental principle of epidemiology and ecology is that the severity and extent of a disease or pest outbreak depends on host population density — and monoculture maximises this density [31]. Within the first 50 years of tilling, 40–70% of the original store of soil organic matter (carbon and nitrogen) is lost. Monoculture systems in Kenya and Bolivia were found to be highly susceptible to climate shocks, price volatility, soil fertility loss, and pest outbreaks — demonstrating the structural fragility of agroindustrial monocropping [32].

The vulnerability inherent in genetic uniformity is not theoretical. The 1970 Southern Corn Leaf Blight — caused by a single cytoplasmic trait shared by virtually all US hybrid maize varieties — destroyed 15% of the American crop in a single season and caused approximately \$1 billion in losses, a direct consequence of the genetic homogeneity that monoculture systems create [33]. The structural driver of this homogeneity is partly economic: two companies (Syngenta Group and Bayer) control 40% of the global commercial seed market, while four firms control 62% of the world agrochemical market — market concentrations that meet the standard economic definition of oligopoly [34]. Even where many farms exist, their effective genetic choices are constrained by a handful of input suppliers, making homogenisation not merely an agronomic preference but the structural outcome of concentrated market power.

2.7 Freshwater depletion

Agriculture is responsible for nearly 70% of global freshwater withdrawal volume, making it the most water-intensive sector by far [35]. As much as 60% of agricultural water is lost due to inappropriate irrigation systems and poor application techniques, contributing to the depletion of rivers, lakes, and underground aquifers.

The full scale of agricultural water consumption is partially obscured because it includes the depletion of non-renewable aquifers. Global groundwater depletion increased 22% in just one decade, from 240 km³/year in 2000 to 292 km³/year in 2010, with five commodity crops — wheat (22%), rice (17%), sugar (7%), cotton (7%), and maize (5%) — accounting for 83% of all groundwater depletion embedded in international food trade [36]. Unlike surface water shortages, depletion of fossil aquifers is functionally irreversible on human timescales: some of the groundwater currently consumed to irrigate these crops fell as precipitation thousands of years ago and will not recharge within any planning horizon relevant to food security policy.

2.8 Greenhouse gas emissions

Global agrifood systems emit approximately 16.2 billion tonnes of CO₂-equivalent per year, accounting for roughly one third of total anthropogenic greenhouse gas emissions [37]. These are generated across farm production, land-use change, and the entire supply chain including processing, retail, and food disposal.

The aggregate figure masks the severity of individual gas contributions. Nitrous oxide from agricultural nitrogen use alone grew 30% between 1980 and 2020, increasing faster than any IPCC scenario had projected [38]. An independent lifecycle analysis places the total food-systems contribution at 34% of all anthropogenic greenhouse gas emissions — 18 Gt CO₂-eq/year as of 2015 — with post-farm supply-chain activities (processing, packaging, transport, retail, and disposal) already on track in many countries to exceed farm-gate and land-use-change emissions as the dominant source [39].

2.9 Produce transportation, storage and disposal

Producing, transporting, and allowing food to rot contributes more than 8–10% of global greenhouse gas emissions — nearly five times the total emissions from the entire aviation sector [40]. Food waste comprises about 24% of municipal solid waste in landfills, and due to its fast decay rate, an estimated 58% of fugitive methane emissions from municipal solid waste landfills are attributable to landfilled food waste [41]. As production efficiency increases, the relative weight of these post-farm emissions grows, making local, short-supply-chain production an increasingly important lever for climate mitigation [39].

2.10 The existential dependency on a few producers

A significant 60% of the world’s food production occurs in just five countries: China, the United States, India, Brazil, and Argentina [42]. This geographic concentration makes a substantial segment of the world’s food supply susceptible to geopolitical disruption, extreme weather, and infrastructure failure. The world has experienced three global food crises in the past 50 years, each exposing the same underlying structural feature: the global industrial food system’s reliance on a small number of staple grains produced using highly

industrialised farming methods, making it susceptible to events affecting just a handful of crops or regions [43].

The concentration risk extends upstream beyond geography. In the commercial seed sector, two companies (Syngenta Group and Bayer) control 40% of the global market; in agrochemicals, four firms control 62%; similar oligopolistic concentration characterises agricultural machinery and animal pharmaceuticals [34]. This upstream concentration creates systemic risk through interlocking ownership structures that reduce effective competition and introduce single points of failure invisible in geographic analyses of production alone [44]. The existential dependency is therefore not merely on a few producing regions, but on a small number of firms that supply the inputs upon which those regions — and the rest of the world’s food supply — depend.

2.11 Suppression of dietary diversity and nutritional quality

Industrialised agriculture prioritises monoculture production, contributing to the global proliferation of ultra-processed foods and the concurrent rise in obesity and non-communicable diseases including diabetes and cardiovascular disease, alongside the drastic acceleration of ecosystem decline [45].

The scale of this homogenisation is empirically documented. An analysis of national food supply data from 152 countries over 50 years found that global diets converged significantly towards a common pattern dominated by wheat, rice, maize, and sugar, with historically diverse regions experiencing the most rapid homogenisation [46]. Modelling by Springmann et al. shows that current dietary trends — dominated by industrial monoculture commodity crops and animal products — will push multiple planetary boundaries beyond safe limits by 2050, including those for climate change, land use, freshwater use, and nitrogen pollution, even under optimistic assumptions about production efficiency improvements; diversification of diets towards plant-rich, varied foods is identified as the single highest-impact lever available to policymakers [47].

3 Background: The Netherlands as a Proof of Concept

3.1 The Dutch agricultural model

The Netherlands is the world’s second largest exporter of agricultural products after the United States, despite having a land area roughly the size of West Virginia [48].¹ This extraordinary productivity is not the result of favourable geography — the Netherlands is a flat, densely populated, low-lying country with no particularly exceptional climate — but of decades of deliberate investment in controlled-environment agriculture, precision management, and a collaborative institutional model.

Approximately two decades ago, the Dutch government and agricultural sector committed to a principle that seemed audacious at the time: to produce twice as much food using half as many resources [49]. Since 2000, Dutch farmers have reduced water dependence for key crops by up to 90%, and chemical pesticide use in greenhouses has been reduced to near zero through replacement by biological control systems [49]. A leading Netherlands greenhouse produces approximately 80 kg of tomatoes per m² per growing season, compared with roughly 4 kg/m² in a comparable field operation in Spain — a twentyfold yield advantage on the same footprint [49].

The water efficiency numbers are equally striking. Producing 1 kg of tomatoes under field conditions in a temperate climate requires approximately 60 litres of water. A standard greenhouse reduces this to around 12.5 litres, and Dutch closed hydroponic systems with recirculating drainage reduce it further to as little as 4 litres per kilogram — a fifteenfold improvement over the field baseline [50]. High-tech greenhouse vegetable production in the Netherlands requires at least 75% less land than comparable field cultivation methods used elsewhere in Europe, and more than 50% less water, with wastewater and nutrients recycled in closed loops [51].²

3.2 The Triple Helix institutional model

The institutional structure behind this performance is as important as the technology. Dutch agricultural innovation has been driven by a sustained collaboration between three partners: the national government (providing policy alignment, long-term research funding, and infrastructure), Wageningen University & Research (providing applied scientific research, education, and agronomic expertise), and the private sector (translating research outputs into commercial systems and equipment). This arrangement — sometimes described as a Triple Helix model of innovation — created a virtuous cycle in which commercial success financed further research, policy rewarded resource efficiency, and knowledge diffused rapidly through grower networks and extension services. The Netherlands holds only 1% of EU farmland yet produces 6% of its food, and its per-unit-of-output performance is the highest in Europe [52].

¹This figure includes re-exported products and ornamental crops; declared export value includes substantial re-export and processing activity. Dutch net value-added agricultural exports nonetheless remain the second largest in the world by most measures.

²The Marcelis et al. (2019) study was commissioned by Grodan, a substrate company. Headline efficiency figures are broadly consistent with independent peer-reviewed sources and the Wageningen University affiliation lends scientific credibility; nonetheless the conflict of interest should be noted.

3.3 Limitations and honest framing

The Dutch model contains real contradictions that must be stated plainly. Despite achieving world-leading per-product efficiency, Dutch agriculture generates 14% of national greenhouse gas emissions [52]. Emissions of ammonia, nitrogen surplus, and pesticide use *per hectare* remain among the highest in the EU despite decades of efficiency gains [52]. The absolute environmental load per unit of land has not decreased in proportion to per-product efficiency gains, because the intensive model concentrates production and inputs at extreme densities. This tension culminated in the 2019–2023 Dutch nitrogen crisis, in which court rulings forced the government to order substantial livestock reductions to comply with EU nitrogen deposition limits, triggering widespread farmer protests and a national political realignment.

High-tech greenhouse production also carries significant energy costs. Year-round enclosed cultivation requires substantial inputs for heating, artificial lighting, and climate control. The resource efficiency gains of the Dutch greenhouse sector are real, but their sustainability depends heavily on the source and cost of that energy — a dependence that must be resolved for the model to be genuinely sustainable at scale.

3.4 Relevance to the proposed system

The Dutch case study establishes two distinct and important findings.

First, it demonstrates empirically that the efficiency improvements attributed to controlled-environment and precision agriculture are not speculative. Resource reductions of 75–90% relative to conventional field agriculture have been achieved and sustained at national scale over more than two decades. The core claim of precision-managed, closed-loop cultivation — that it can produce substantially more food from substantially less land, water, and chemical input — is not a hypothesis. It is a documented outcome.

Second, it reveals where the current model falls short. The Dutch system is dependent on concentrated capital, industrial-scale infrastructure, proprietary technology, and a uniquely well-resourced institutional ecosystem that most of the world cannot replicate. The environmental gains are real, but they are locked inside a commercial model that is inaccessible to smallholder farmers, low-income communities, and regions that lack the capital or institutional infrastructure to participate. The efficiency is real; the access is not.

The system proposed in this paper extends the proven technical core of the Dutch approach — closed-loop nutrient cycling, sensor-guided precision management, and controlled-environment cultivation — into a framework that is open-source, modular, and explicitly designed for deployment by communities with minimal institutional support or capital expenditure. It takes what the Dutch model has demonstrated to be achievable and asks how to make it available to everyone.

4 The proposed solution

4.1 Concept overview

The platform is a small-scale, open-source, sensor-driven greenhouse with the following operational loop:

1. A sensor array takes periodic measurements of all relevant environmental variables (temperature, relative humidity, CO₂ concentration, pH, electrical conductivity, and light intensity).
2. A camera periodically images the crop. A computer vision model extracts growth metrics, identifies disease symptoms, and can identify species.
3. An ML growth model — trained on the shared dataset and running inference at the edge — computes optimal environmental set-points to maximise yield while minimising resource consumption for the current species and growth stage.
4. A PID controller on a microcontroller drives the actuators (LED grow lights, circulation fans, nutrient dosing pumps, heating/cooling elements) to reach and maintain the computed set-points.
5. Anonymised telemetry is periodically uploaded to a shared data commons, where it is used to retrain models, which are then pushed back to all nodes via over-the-air updates.
6. A climate-aware species recommendation engine suggests crops whose optimal conditions best match the local ambient environment.

All hardware, firmware, software, models, and data are open source.

4.2 Alignment with identified problems

The platform directly addresses the problems identified in the preceding section:

Problem	Mechanism	Type of impact
Land use / habitat loss	CEA yields 10–100× more per unit of land area than field farming	Direct
Biodiversity loss	Reduced land pressure frees habitat; proximity reduces monoculture incentive	Direct
Soil erosion and degradation	Soilless operation entirely bypasses tillage-driven erosion	Bypass
Fertiliser and eutrophication	Closed nutrient loop with near-zero runoff	Direct
Pesticide pollution	Enclosed environment is a physical pest barrier; greatly reduced need	Direct
Monoculture	Platform enables diverse crop selection per node	Enables diversity
Freshwater depletion	Hydroponic/aeroponic systems use 70–95% less water than field agriculture	Direct
GHG emissions	Reduced transport, waste, and land conversion; conditional on energy source	Conditional
Transport and food waste	Proximity to consumers reduces both transport emissions and spoilage	Direct
Supply-chain concentration	Network of many small producers is structurally more resilient	Direct
Dietary diversity	Each node can grow different crops, supporting variety the industrial system cannot	Direct

The platform introduces one new risk not present in field agriculture: **energy intensity**. Controlled environment agriculture trades land and water efficiency for electrical energy consumption. Whether this trade is environmentally beneficial depends on the local energy mix. This is addressed in Section 6.

4.3 Scope and niche

The platform is not designed to replace all agriculture. It is designed for a specific, well-defined niche: **high-value, water-intensive, perishable crops grown in proximity to consumers**. This niche includes leafy greens, herbs, tomatoes, cucumbers, strawberries, peppers, and similar produce — crops for which the

transport, water, and pesticide savings of CEA are largest, and for which the calorie-per-watt ratio is most favourable relative to field alternatives.

The platform does not target caloric staples (grains, legumes at scale) where field agriculture remains more energy-efficient per kilocalorie. However, it can meaningfully contribute to nutritional self-sufficiency at the household level, particularly when the species recommendation engine selects crops that optimise simultaneously for nutritional completeness and energy efficiency.

5 Nutritional self-sufficiency analysis

5.1 The calorie problem

Full caloric independence from a small greenhouse is not achievable for most people with the crops CEA does best. The caloric yield per unit area of the typical CEA crop profile is too low [53]:

Crop	Yield (kcal/m ² /year)	Area for one adult (m ²)
Lettuce	300	2,400
Spinach	500	1,460
Tomato	3,000	243
Dry beans	4,000	183
Potato	7,000	104
Sweet potato	8,000	91

An adult requires approximately 730,000 kcal/year. A mixed CEA system optimised for calorie-dense crops (sweet potato, potato, dry beans) could cover one adult’s caloric needs in roughly 100–200 m² of greenhouse space — feasible for a large backyard installation but not a typical small greenhouse.

5.2 The correct framing: nutritional self-sufficiency

The appropriate goal is not caloric independence but **nutritional self-sufficiency** — specifically, the ability to produce the full spectrum of vitamins, minerals, and phytonutrients needed for health, with meaningful caloric supplementation from a small number of well-chosen calorie-dense crops.

This framing is both more honest and more compelling. It is the micronutrient and phytonutrient density of fresh produce that the industrial supply chain does worst at preserving — nutrients degrade during the long transport and storage times that are inherent to centralised production. A locally grown, freshly harvested crop is nutritionally superior to the same crop shipped across a continent.

The species recommendation engine should therefore optimise for **nutritional completeness** as a primary objective, with caloric contribution and energy cost as co-objectives. This makes the goal achievable in realistic backyard footprints for the majority of a person’s fresh produce needs.

5.3 Vertical stacking multiplier

CEA systems can stack crops vertically, multiplying the effective growing area per unit of floor space by 10–20×. This means the effective growing area available in even a modest installation can substantially exceed its floor footprint, making the nutritional self-sufficiency target considerably more achievable than the raw floor-area numbers suggest.

6 Limitations and mitigations

6.1 Energy intensity

The dominant weakness of controlled environment agriculture is electrical energy consumption. Lighting, HVAC, and humidity control all require electricity. If that electricity is generated from fossil fuels, the environmental benefit from reduced land use and water consumption may be partially or fully offset by increased carbon emissions.

Mitigation strategy:

- The species recommendation engine directly addresses this by recommending crops whose optimal growing conditions minimise the delta between ambient environment and required conditions — thereby minimising heating, cooling, and supplemental lighting needs.
- The system monitors and logs energy consumption per growth cycle, enabling users to make informed decisions about energy sourcing.
- The platform is designed to integrate with solar photovoltaic systems. Where local solar generation is available, the system can shift energy-intensive operations (lighting, active climate control) to periods of surplus generation.
- The whitepaper explicitly scopes the platform to crops where the environmental tradeoffs favour CEA even with grid electricity — primarily water-intensive crops where the water savings alone justify the energy overhead.

The energy problem is not fully solved by this platform. In high-latitude climates with short winter days and fossil-fuel-dominated grids, the environmental case for fully artificial-light CEA is weak. The platform should be honest about this in its documentation and recommendation engine output.

6.2 Cold-start problem for the ML model

The growth optimisation model requires training data to be useful, but early deployments have no training data. The flywheel only works once the network is large enough to produce statistically useful data.

Mitigation: The initial model is bootstrapped from existing published datasets (PlantVillage for CV; published hydroponic growth studies for the environmental optimisation model). The system is useful from day one — it provides automated control and monitoring — even before the ML model reaches meaningful optimisation capability. Early adopters are incentivised by automation, not optimisation.

6.3 Open-source hardware complexity

Open-source hardware is harder to distribute than software. Someone must fabricate components, assemble systems, and calibrate sensors.

Mitigation: A tiered hardware support model is proposed (see Section 8.3). The project maintains a small number of fully validated reference configurations. Community-contributed BSPs and hardware variants are documented separately, with clear statements about their validation status.

6.4 Data governance and quality

A shared dataset is only as useful as the quality and consistency of the data it contains. Different sensor configurations, calibration procedures, and tagging conventions will degrade model quality if not managed.

Mitigation: The data schema mandates a fixed set of tags for every telemetry record: `node_id` (hashed), `species_id` (from GBIF taxonomy), `climate_zone` (Köppen classification), `growth_stage`, and `hardware_config` hash. The ingest service enforces schema validation before any record enters the commons. A data quality score is computed per node and used to weight training samples.

7 System architecture

7.1 Overview

The system has three physical tiers:

1. **Greenhouse node** — embedded hardware (MCU + SBC) inside the greenhouse.
2. **Edge gateway** — the SBC itself, running Linux, hosting the local API, data store, and edge ML runtime.
3. **Cloud commons** — optional, self-hostable, handling data aggregation, model training, and OTA distribution.

The node and gateway operate fully offline. The cloud connection is asynchronous and optional. All user data is anonymised before leaving the local network.

7.2 The greenhouse node: MCU/SBC split

The node contains two levels of compute with fundamentally different requirements.

MCU subsystem (hard real-time):

- Sensor polling at configurable intervals (temperature, humidity, CO₂, pH, EC, light).
- PID control loop — continuously running, must not be interrupted.
- Actuator drivers (PWM for LED lights, relay control for pumps and fans).
- Set-point receiver: accepts target values from the SBC via UART/I2C.
- Safety watchdog: if the SBC goes silent, holds actuators at last known safe set-points.

SBC subsystem (latency-tolerant, Linux):

- Camera capture and CV inference pipeline.
- Edge ML runtime (ONNX Runtime): runs the growth model and produces set-point recommendations.
- Set-point planner: translates model output into PID target values.
- Local time-series store: ring buffer of all sensor readings and camera metadata.
- Local REST API: serves the user dashboard and receives configuration.
- Sync agent: handles asynchronous telemetry upload and OTA model download.

The MCU/SBC split is a deliberate safety design. A kernel panic, a slow CV inference job, or an OTA update on the SBC cannot destabilise the physical environment — the MCU continues running the PID loop with its last-known set-points until the SBC recovers.

7.3 The data flywheel

The cloud commons implements the learning loop:

1. Deployed nodes periodically upload anonymised telemetry (sensor readings, growth stage, yield estimates, energy consumption) to the ingest service.
2. The ingest service validates schema, anonymises any remaining identifiers, and writes records to the time-series data store and image object store.
3. A training pipeline continuously retrains the growth model and CV model on the growing dataset.
4. Trained models are published to the model registry with version numbers.
5. The OTA update service pushes new model versions to nodes, which download and apply them without interrupting operation.

The species recommender is not an ML model. It is a constrained optimisation over a structured database of crop requirements, filtered by local climate data and energy cost estimates. This makes it interpretable and auditable without needing training data.

7.4 Offline-first principle

The system is designed to be fully functional without any internet connection. A user who never connects to the cloud commons loses only the improving models and the species recommendation service — all monitoring, control, and local data logging work offline. This is essential for the project’s goal of being deployable anywhere, including locations with poor connectivity.

8 Technology stack

8.1 Embedded firmware (MCU)

Concern	Choice	Rationale
Language	C / C++ (Rust viable for safety-critical paths)	Non-negotiable for bare-metal MCU targets
RTOS	FreeRTOS (primary); Zephyr RTOS (alternative)	FreeRTOS: mature, MIT licence. Zephyr: better multi-arch HAL abstraction
PID implementation	Custom, runtime-configurable gains	Simple enough to build and test in-house; avoids unnecessary dependencies
Serial protocol	UART/I2C with COBS framing	Consistent Overhead Byte Stuffing gives reliable framing with minimal overhead
Multi-arch build	CMake with per-target toolchain files	Supports ESP32, RP2040, STM32 from a single codebase via platform abstraction layer

8.2 Edge gateway (SBC, Linux)

Concern	Choice	Rationale
OS build system	Yocto Project	Reproducible, minimal embedded Linux; multiple BSPs as separate Yocto layers
Edge ML runtime	ONNX Runtime	Best cross-architecture support (x86, ARM64, ARM32); accepts PyTorch and TensorFlow exports
CV framework	OpenCV + ONNX Runtime	OpenCV handles capture and preprocessing; ONNX Runtime runs model inference
Local time-series store	InfluxDB (or SQLite ring buffer)	InfluxDB: purpose-built for sensor telemetry. SQLite: lighter alternative for constrained hardware
MCU-SBC IPC	Serial + Protobuf or MessagePack	Versioned binary schema allows both sides to evolve independently
Local API	FastAPI (Python) or Axum (Rust)	FastAPI: quick development. Axum: lower memory overhead for constrained SBCs
Local dashboard	React or HTMX	HTMX preferred: avoids JS build toolchain on a device users flash, not develop on
OTA update client	SWUpdate or RAUC	Mature A/B update frameworks for embedded Linux; safe roll-back on failure

8.3 Hardware support tiers

To manage the complexity of multi-platform support without overcommitting the core team:

- **Tier 1 (fully supported):** 2–3 reference hardware configurations validated by the core team on every release. Failures here block release.
- **Tier 2 (community supported):** Additional BSPs and hardware variants maintained by contributors. Best-effort validation. Status clearly documented.

This mirrors the approach used by mature embedded Linux projects such as OpenWRT and Buildroot.

8.4 Cloud / data commons

Concern	Choice	Rationale
Language	Python	Standard for ML pipelines
ML framework	PyTorch	Dominant in research; good ONNX export support
Growth model	Gaussian Process regression → neural surrogate	GP: interpretable, works with small datasets, provides uncertainty estimates. Transition to neural surrogate as data volume grows
CV model	EfficientNet-B0 or MobileNetV3 (fine-tuned)	Small enough to run on SBC; fine-tuned on PlantVillage + community dataset
Species recommender	Constrained optimisation (scipy / OR-Tools)	Not ML — structured search over a crop requirements database. More interpretable, no training data needed
Data ingestion	MQTT broker → TimescaleDB (or Kafka for high volume)	TimescaleDB is a PostgreSQL extension; familiar to most contributors
Image storage	MinIO (self-hostable, S3-compatible)	Critical for open-source self-hosting
Model registry	MLflow	Tracks experiments, stores model versions, provides serving API
CI/CD	GitHub Actions + hardware-in-loop tests	Firmware CI builds all Tier 1 targets on every PR; at least one physical test node in CI
Hosting	Fly.io, Railway, or Hetzner VPS	Avoid cloud provider lock-in; entire stack should be docker compose up-able

8.5 Licences

Asset	Licence
Firmware and software	AGPL-3.0
Hardware reference designs	CERN OHL-S v2
Trained models	Apache 2.0
Dataset	CC-BY 4.0
Documentation	CC-BY 4.0

AGPL-3.0 for software ensures that cloud forks must open-source their modifications. CERN OHL-S is the standard for open hardware. Apache 2.0 for models allows broad use while requiring attribution. CC-BY 4.0 for data and documentation maximises reusability.

9 The data schema

Every telemetry record must carry the following mandatory fields. This schema is enforced by the ingest service and is version-controlled.

9.1 Sensor telemetry record

Field	Type	Description
<code>node_id</code>	string (SHA-256 hash)	Anonymised node identifier
<code>timestamp</code>	ISO 8601 UTC	Measurement time
<code>species_id</code>	string (GBIF taxon key)	Crop species being grown
<code>climate_zone</code>	string (Köppen)	Local climate classification
<code>growth_stage</code>	enum	seedling / vegetative / flowering / fruiting / harvest
<code>hardware_config</code>	string (hash)	Hash of sensor and actuator configuration
<code>temperature_c</code>	float	Air temperature (°C)
<code>humidity_pct</code>	float	Relative humidity (%)
<code>co2_ppm</code>	float	CO ₂ concentration (ppm)
<code>ph</code>	float	Nutrient solution pH
<code>ec_ms_cm</code>	float	Electrical conductivity (mS/cm)
<code>light_mol_m2_d</code>	float	Daily Light Integral (mol/m ² /day)
<code>energy_kwh</code>	float	Energy consumed since last record
<code>actuator_states</code>	JSON object	Current state of all actuators
<code>set_points</code>	JSON object	Current PID target values

9.2 Image record

Field	Type	Description
image_id	UUID	Unique image identifier
node_id	string (hash)	Linked node
timestamp	ISO 8601 UTC	Capture time
species_id	string (GBIF)	Crop species
growth_stage	enum	As above
bounding_boxes	JSON array	Per-plant bounding boxes where available
cv_annotations	JSON object	Model outputs: growth score, disease flags, confidence
image_path	string	Path in object store

10 The climate-aware species recommender

10.1 Concept

The species recommender is the component that addresses the energy problem most directly. Rather than specifying a fixed set of crops and conditioning the environment to suit them, it inverts the relationship: given a local climate profile, it finds the crops whose requirements best match what the environment already provides, minimising the energy needed to close the gap.

10.2 Input variables

- Local ambient temperature profile (monthly min/max/mean).
- Local humidity profile.
- Local solar radiation / DLI profile.
- Greenhouse construction type (passive solar / supplemental lighting / full artificial).
- Energy source and cost (grid / solar / mixed).
- Available floor area and stacking capability.
- Nutritional objectives (user-specified or defaulting to WHO micronutrient completeness).

10.3 Output

For a given configuration, the recommender outputs a ranked list of species with:

- Estimated energy cost per kg of yield.
- Estimated yield per m² per year.

- Nutritional contribution score.
- Optimal growing season (or year-round if climate allows).
- Suggested crop rotation or companion planting combinations.

10.4 Why this is not an ML model

The recommender is implemented as a constrained optimisation over a structured crop database, not as a learned model. This is a deliberate choice: the decision is deterministic and auditable, the database of crop requirements can be curated and cited from agronomic literature, and the system produces useful recommendations from day one without needing training data.

The database is itself open and versioned, enabling community contributions and corrections. Species entries include: optimal temperature range, optimal humidity range, minimum DLI, nutrient requirements, typical yield per m²/year under CEA, water requirement, and days to harvest.

10.5 Climate adaptation over time

As local climates shift, the recommender naturally recalibrates — suggesting different species as the ambient temperature envelope changes. This gives the platform a long-term climate resilience property: it adapts to a changing environment rather than becoming obsolete as growing conditions shift.

11 Open-source strategy

11.1 The data commons as the core asset

The most valuable long-term output of the project may not be the hardware design or even the software — it may be the **shared open dataset**. A large, well-structured, openly licensed dataset of plant growth under controlled conditions, tagged with species, sensor readings, yield outcomes, disease events, and energy consumption, would be a genuinely novel resource. Nothing comparable currently exists for small-scale CEA at the level of granularity this system produces.

This dataset has value to researchers, plant breeders, agronomists, and other engineers independently of any single product. It should be treated as a first-class project output, not a by-product.

11.2 Community governance

The project should establish governance structures from the outset to prevent the data commons from becoming a de-facto proprietary asset:

- The dataset is hosted on infrastructure that the community can mirror.
- Model weights are published with provenance (which dataset version, which training run).
- The species database is a separately maintained community project.
- Ingest schema changes require a deprecation period and community discussion.

11.3 Self-hostability

The entire cloud stack must be deployable by any community member with a modest VPS. The deployment target is `docker compose up` from a single repository. This prevents the project from creating a dependency on a single hosted service and ensures continuity if the founding contributors step back.

12 Impact assessment

The following is a summary evaluation of the project’s potential impact across the eleven problems identified in Section 2.

Problem	Impact type	Confidence	Key condition
Land use / habitat loss	Direct, strong	High	Deployment reaches meaningful scale
Biodiversity loss	Direct, strong	High	As above
Soil erosion	Bypass (soilless)	High	Not a remediation of damaged land
Fertiliser / eutrophication	Direct, strong	High	Closed-loop nutrient management implemented correctly
Pesticide pollution	Direct, strong	High	Enclosed environment maintained; fungal risks managed
Monoculture	Enables diversity	Medium	Depends on user and community crop choices
Freshwater depletion	Direct, strong	High	Hydroponic/aeroponic systems required
GHG emissions	Conditional	Medium	Conditional on low-carbon electricity
Transport / food waste	Direct, strong	High	Local deployment required
Supply-chain concentration	Direct, strong	High	Deployment breadth required
Dietary diversity	Direct	Medium	Species recommender + user education
Energy intensity (new risk)	Risk introduced	High	Mitigated by recommender; not eliminated

13 Future work

- Integration with home energy management systems (excess solar $\rightarrow$ priority grow-light scheduling).
- Federated learning implementation to allow model improvement without raw data leaving the node.
- Aquaponics extension: closed-loop fish + plant system for protein production.
- Seed library integration: linking the species database to open seed repositories.
- Economic model: cost-per-kg analysis across species, climate zones, and hardware configurations to help users understand the financial viability of their setup.

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